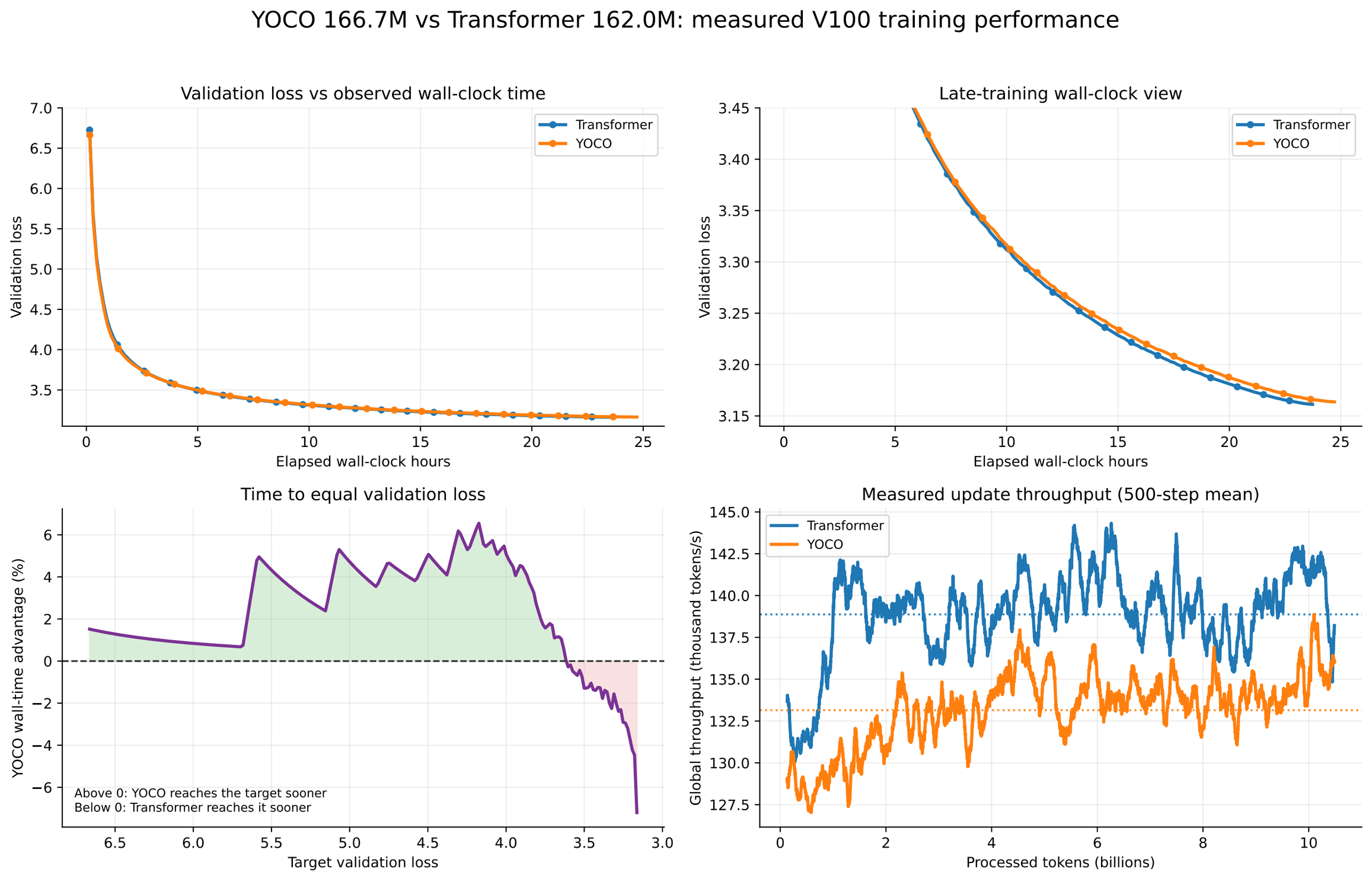

YOCO 166.7M vs Transformer 162.0M: measured V100 training performance
Validation loss vs observed wall-clock time
Validation loss
Elapsed wall-clock hours
Transformer
YOCO
Late-training wall-clock view
Validation loss
Elapsed wall-clock hours
Transformer
YOCO
Time to equal validation loss
YOCO wall-time advantage (%)
Above 0: YOCO reaches the target sooner
Below 0: Transformer reaches it sooner
Target validation loss
Measured update throughput (500-step mean)
Global throughput (thousand tokens/s)
Transformer
YOCO
Processed tokens (billions)
Directly observed on the paired 32×V100 runs; no 6ND axis. Train start→final validation: 23.73 h Transformer, 24.71 h YOCO (1.041×). YOCO peak memory is 19.1% lower. Dotted throughput lines show full-run means.